Niche overlap in Species Distribution Modeling

Modeling the distribution of marine biodiversity under global climate change

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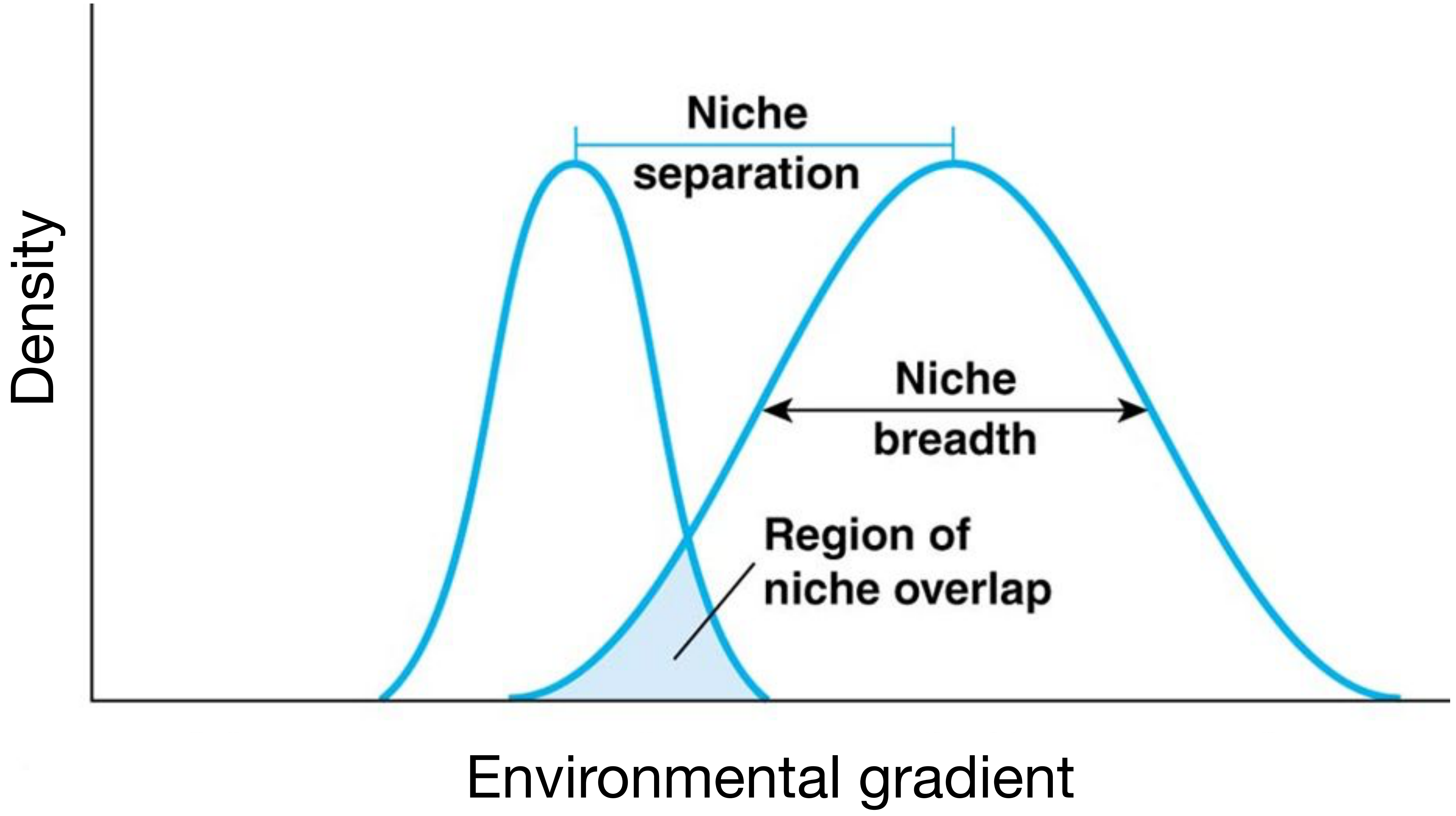


Niche overlap

Niche overlap: the degree to which two species use the same environmental conditions or resources - how much their ecological requirements coincide.

Allow researchers to quantify and compare environmental niches.

Why addressing niche overlap?

Understanding species coexistence and competition: high niche overlap where they co-occur can suggest strong high interspecific competition.

Investigating speciation and niche evolution: comparing niches can provide insights into speciation processes.

> **e.g., sister species:** did species diverge into different environmental niches (ecological speciation), or do they retain similar niches (niche conservatism), and speciated via other mechanisms (like geographic isolation - allopatry)?
>
> **e.g., invasive species:** comparing the niche of an invasive species in its native versus invaded range can reveal whether the species has shifted its niche to succeed in the new environment or not (niche conservatism).

(…)

Approaches to measuring niche overlap

Geographic overlap (e.g., distribution map comparison);

Physiological response overlap (e.g., Laboratory or SDM functions);

Environmental overlap (environmental space).

Environmental overlap

Indices to measure the degree of niche overlap - quantitative, standardized ways to measure how much two species' niches overlap - based on their environmental distributions. How much of the "niche density" overlaps?

Schoener's D: gives more weight given to high-probability areas - places where species are most likely to be - focuses on the amount of shared niche space.

Hellinger's I: transforms probabilities using a square root, reducing the influence of extreme values (like very high suitability) - **softer measure** that gives more weight to overlap when suitability values aren't extreme.

Both D and I provide a value **between 0 (no overlap) and 1 (complete overlap)**, making it easy to interpret and compare.

Environmental overlap

Equivalency test: are the species using the same niche?

Similarity test: are the species' niches more alike than expected by chance?

If $p < 0.05$, the difference (or similarity) is unlikely due to chance (i.e., different niches).

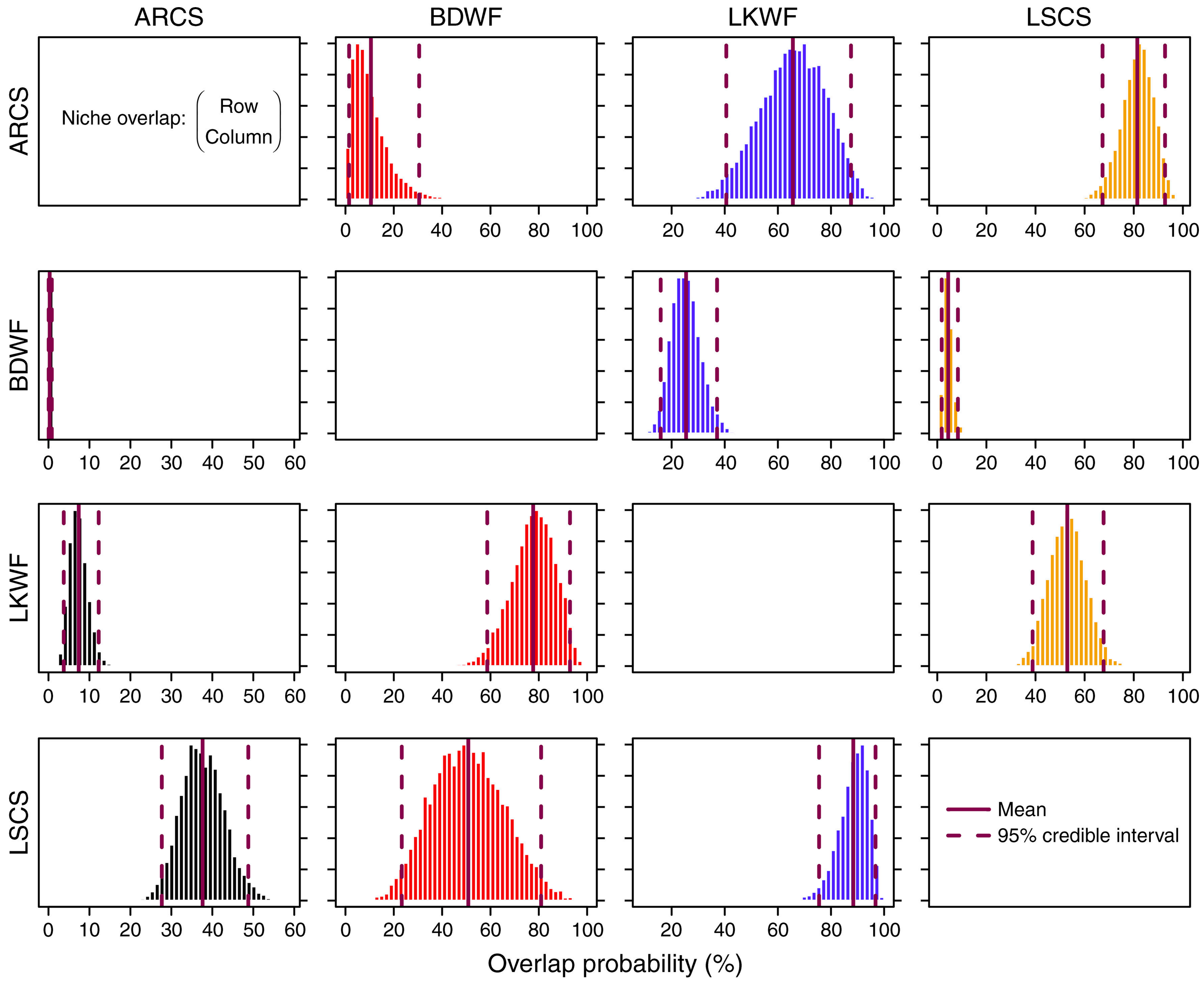
If $p > 0.05$, the observed overlap is within the range of random expectation.

Probabilistic method for quantifying n-dimensional niches

Defines overlap as the probability that individuals from species A occur in B’s region.

Directional, multivariate, Bayesian framework.

Consistent with Hutchinson’s n-dimensional hypervolume.



Distribution of the probabilistic niche overlap metric (%) for a specified niche region - **probability of species displayed in rows overlapping onto those displayed in columns**. Mean and 95% intervals of niche overlap in purple.

References

Swanson et al., 2015: https://doi.org/10.1890/14-0235.1

Schoener, 1968: https://doi.org/10.2307/1935534

Warren et al., 2018: https://doi.org/10.1111/j.1558-5646.2008.00482.x